MIL-E-1/904D
7 February 1972
SUPERSEDING
MIL-E-1/904C
18 November 1969

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6197

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Power pentode

Outline --- 6-3 (EIA)

Base --- E9-1

Envelope --- T6-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	k	g1	g2	h	h	a	g3 int sd	g2	g1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	(D) Ef	Eb	Ec1	Ec2	Shield & Ec3	Pp	Pg2	Ehk	Ik	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	W	W	v	mAdc	°C	ft
Maximum:	6.6	300	0	250	0	7.5	2.5	180	50	200	Note 1
Minimum:	6.0	---	-50	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	250	-3.0	150	0	---	---	---	---	---	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)	-		0.65	II	Ib	20	40	mAdc
1256	Electrode current (2) (anode)	-	Eb = 50 Vdc; Ec1 = 0; Ec2 = 100 Vdc	0.65	II	Ib	26	46	mAdc
1256	Electrode current (3) (anode)	-	Ec1 = -12 Vdc	0.65	II	Ib	---	100	µAdc
(D) 1266	Total grid current	2	Rg1 = 0.25 Meg	0.65	II	Ic1	---	-1.5	µAdc
1256	Electrode current (screen)	-		0.65	II	Ic2	5.0	9.0	mAdc
1306	Transconductance (1)	-		0.65	II	Sm	9,000	13,000	µmhos
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1256	Electrode current (4) (anode)	-	Ebb = 150 Vdc; Ec1 = 0; Ec2 = 90 Vdc; Rb = 3,000 ohms; Rg2 = 470 ohms	6.5	Code E	Ib	20	36	mAdc
(D) 1256	Electrode current (5) (anode)	3, Fig 1	Eb = 250 Vdc; Ec1 = 0; Ec2 = 150 Vdc; Rg1 = 0.47 Meg	1.0	II	ib	28	---	ma
1301	Heater current	-		6.5	Code E	If	610	690	mAdc
1336	Heater-cathode leakage	-		2.5	I	Ihk	---	40	µAdc
1306	Transconductance (2)	-	Ef = 5.7 V	6.5	Code E	Sm	8,500	---	µmhos
---	Transconductance (screen)	-		6.5	Code E	Sm (g1-g2)	1,700	2,900	µmhos
1246	Audio frequency noise	-	Esig = 150 mVac; Eb = Ec2 = 225 Vdc; Rg1 = 0.25 Meg; Rk = 150 ohms; Rg2 = 5,000 ohms; Rb = 10,000 ohms	6.5	Code E	EB	---	17	vu
1331	Direct-interelectrode capacitance	-		6.5	Code E	Cg1p Cin Cout	--- 9.2 4.0	0.135 13.8 6.0	pF pF pF
1316	Amplification factor (triode)	-	Eb = Ec2 = 150 Vdc (g2 tied to anode)	6.5	Code E	Mu	19	25	---
---	Intermittent shorts	5, 6		2.5	I	---	---	---	---
1211	Insulation of electrodes	-		2.5	S4	R	100	---	Meg

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
(D) 1121	Base strain	-		---	---	---	---	---	---
(D) 2126	Glass strain	-		---	---	---	---	---	---
(D) 1105	Permanence of marking	-		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC 1ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3										
1506	Heater-cycling life	-	Ef = 7.5 V; Ehk = 180 Vdc; Ec1 = Eb= Ec2 = 0; 1 min on, 4 min off	---	---	---	---	---	---	---	---
1521	Survival-rate life	-	Intermittent life-test (2), or equivalent conditions	---	---	---	---	---	---	---	---
---	Survival-rate life-test end point:										
1201	Short and discontinuity detection	-		---	---	---	---	---	---	---	---
1501	Intermittent life (1)	7	Eb = 300 Vdc; Ec2 = 150 Vdc; Ec1 = -50 Vdc; Ehk = 100 Vdc; Rg1 = 0.25 Meg; TA = room	---	---	---	---	---	---	---	---
1501	Intermittent life (2)	7	Eb = 250 Vdc; Ec2 = 250 Vdc; Ec1 = -8 Vdc; Ehk = 100 Vdc; Rg1 = 0.25 Meg; TA = room	---	---	---	---	---	---	---	---
---	Intermittent life-tests (1) and (2) end points (1,000 hours):	8									
1211	Insulation of electrodes	-		---	---	2	4	R	50	---	Meg
---	Intermittent shorts	-		---	---	2	4	---	---	---	---
1201	Short and discontinuity detection	-		---	---	1	2	---	---	---	---
---	Cathode interface	9		---	---	2	4	Ri	---	25	Ohms
1256	Electrode current (1) (anode)	-		---	---	2	4	Ib	18	---	mAdc
1266	Total grid current	-		---	---	2	4	Ic1	---	-2.0	µAdc
1336	Heater-cathode leakage	-		---	---	2	4	Ihk	---	60	µAdc
---	Combined defectives	-		---	---	5	10	---	---	---	---
	Periodic-check test										
1031	Low-frequency vibration	4	Rb = 2,000 ohms	---	---	---	---	Ep	---	1,000	mVac

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.

2. This test to be performed at the conclusion of the holding period.

(D) 3. Grid No. 1 drive shall be 30 volts, peak-to-peak, square wave (50-percent duty cycle), 6,000 prr. Read ib on a peak reading instrument.

4. Tubes shall be vibrated perpendicular to a plane through pins 2 and 7.

5. Test on a dc thyratron shorts set, or equivalent. Minimum sensitivity shall be as follows:

   2 K for short lasting 2 μs.
   100 K for short lasting 15 μs.
   2.5 Meg for short lasting 1,000 μs.

   Tubes shall not be rejected for heater-cathode shorts on this test.

6. Tap each tube six times - three times in each of two planes using a felt tapper consisting of a felt mallet head 1.0 inch (25.4 mm) in diameter and 1.0 (25.4 mm) long with 1/4 inch (6.35 mm) concentric hole securely glued flush with end of an 8 by 1/4 inch (203.2 by 6.35 mm) plastic rod. Weight of felt shall be approximately 3.0 grams.

7. For intermittent life tests (1) and (2), the sample size shall be 10 tubes per lot.

8. For acceptance purposes, the results of intermittent life tests (1) and (2) are combined as one 20-tube sample. The acceptance numbers given apply to this total sample. In the event of life-test failures, a second sample of 10 tubes shall be run for each life test, and the resulting 40-tube total sample evaluated to the combined sample acceptance numbers.

9. Test conditions: Ef = 5.7 V; Eb = 100 Vdc; Ib + Ic2 + Ic3 = 1.5 mAdc. Connect g2 and g3 to anode. Read at 500 and 1,000 hours only.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI
Air Force - 11
DSA - ES

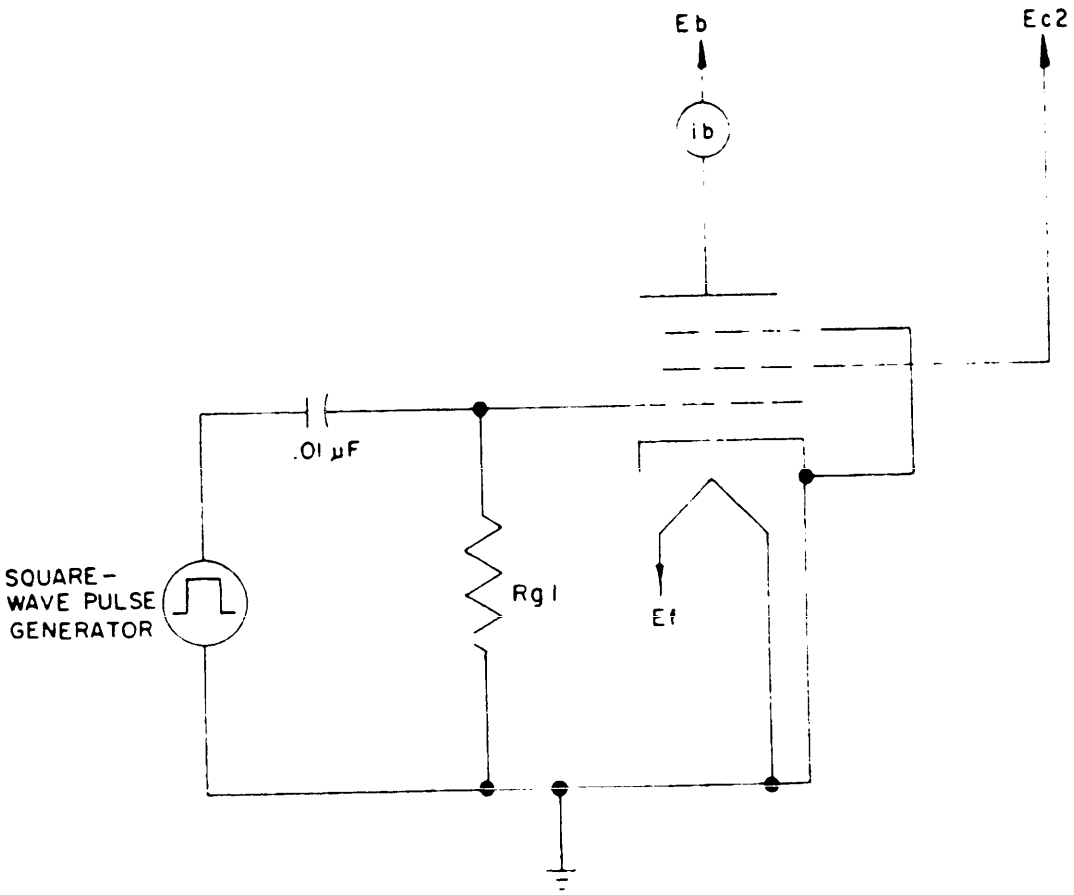
User activities:
Army - ME, WC, AV
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2609)

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(D) FIGURE 1. Peak anode current test with square-wave drive.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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